Research Plan

Daniel Patrick Atwater

November 7, 2021

Contents

Thesis Title 2

Supervisors & Advisors 2

Background 2

Rationale 3

Objective 6

Methodology 6

Outputs 8

Timeline 9

Budget 11

Ethical Approcals 11

WH&S 12

IP 12

Project Infrastructure 12

Skills 12

References 13

Thesis Title

Prognostic landfast sea ice and quantification of the relationship with Antarctic Bottom Water formation off the coast of Cape Darnley, East Antarctica.

Supervisors & Advisors

- ALEX FRASER → Primary Supervisor → AAPP & IMAS
- WILL HOBBS → Supervisor → AAPP & IMAS
- WILMA HUNEKE → Supervisor → ANU
- MAX NIKURASHIN → Grduate Research Coordinator → IMAS

Background

The meridional overturning circulation (MOC) within the Earth's ocean basins is understood to play a significant contribution in regulating heat content of the climate, ventilating the deep ocean and CO_2 uptake (Orsi et al., 1999; Levitus et al., 2005; Bindoff et al., 2007; Purkey and Johnson, 2010; Talley, 2013). With overturning time scales on the order of a millennia (Johnson, 2008) understanding the driving mechanisms of the MOC has implications for the rates of change of key parameters in Earth's climate: e.g. CO_2 uptake rate, latent heat draw down, and ventilation of the deep ocean (Bindoff et al., 2007; Purkey and Johnson, 2010).

The conceptual model of the MOC includes the tropical waters of the Atlantic flowing north via the Gulf Stream and overturning at various locations in the north Atlantic due to baroclinic instabilities, thus becoming North Atlantic Deep Water. At the other end of planet, along the coast of Antarctica, a complex set of seasonal events give rise to coastal processes that cause cold dense water to form at the surface at key locations (Orsi et al., 1999; Talley, 2013). As the water forms it is highly unstable and begins to sink flowing down the slopes of the continental shelf, mixing with ambient waters in its descent, and in the process becoming dense shelf water (DSW) (Orsi et al., 2001). This DSW flows across the Antarctic slope front, with some indication of mixing (Mizuta et al., 2021; Van Achter et al., 2021; Rintoul, 2018), and into the various abyssal plains (e.g. becoming Antarctic Bottom Water —AABW) (Stewart et al., 2019). This water mass fills the ocean basins of the Southern Hemisphere (Johnson, 2008), and is identified as cold dense abyssal water originating from overturning coastal waters around Antarctica (Orsi et al., 1999).

In this manner, AABW mass accounts for 30–40% of global ocean mass (Johnson, 2008). Its formation and controlling mechanisms are areas of contemporary research. In the effort to prognosticate furture climate, the current capability to constrain the climate's heat budget, via global climate models, is dependent upon knowledge of these controlling mechanisms of the deep ocean (Domingues et al., 2008; Murphy et al., 2009; Purkey and Johnson, 2012; Durack et al., 2018). At present, the uncertainty in deep-ocean heat uptake has been shown to be a significant cause of variation among climate models (Boé et al., 2009; Durack et al., 2018). While on the other hand, empirically, it also has been shown that Earth's systems linked to the long-term

water-phase-mass distribution and surface temperature variations are accelerating (Murphy et al., 2009; Pörtner et al., 2019).

One specific uncertainty of the MOC under investigation in this work, is the rate at which AABW is formed via DSW formation. To investigate this uncertainty I will be undertaking a series of simulations of a coupled physical system (sea ice-ocean) over a region of East Antarctica in the vicinity of the Mawson Coast, see 1.

Rationale

The region of East Antarctica under investigation here, is chosen due to its:

1. significance in AABW formation (Williams et al., 2010; Ohshima et al., 2016; Tamura et al., 2016; Williams et al.; Fraser et al., 2019; Aoki et al., 2020),
2. previous remote sensing, oceanographic a geographic studies (i.e. data availability) (Meijers et al., 2010; Williams et al., 2010; Ohshima et al., 2013; Tamura et al., 2016; Fraser et al., 2019; Aoki et al., 2020; Smith et al., 2020), and
3. previous geophysical fluid modelling covering or adjacent to this region and investigating oceanographic results (Galton-Fenzi et al., 2012; Nakayama et al., 2014; Liu et al., 2017; Kusahara et al., 2017a,b; Naughten et al., 2018; Van Achter et al., 2021).

The distinction of the Mawson Coast, in this context, is its inclusion of the Cape Darnley Polynya (CDP). The CDP (Meijers et al., 2010; Ohshima et al., 2013) is a coastal ocean polynya (i.e. latent–heat polynya) (Massom et al., 1998; Tamura et al., 2008). Some coastal ocean polynyas are sources of AABW (Massom et al., 1998; Morales Maqueda et al., 2004; Tamura et al., 2008). For reasons of clarification, I will use the Barber and Massom (2007) definition of polynyas:

> Polynyas are persistent reoccurring regions of open water and/or thin ice, tens to tens of thousands of square kilometres in extent, that occur at locations where more consolidated and thicker ice cover would be climatologically expected.

This defintion is derived, in part, from Smith et al. (1990) and requires a further, modern qualification, and that is that polynyas are *non-linear* regions (Massom et al., 1998) of open water and/or thin ice. The inverse of this given by Smith et al. (1990) in their definition of oceanographic *leads* as *linear* openings in sea ice differentiated by their absence of re-occurrence at a particular geographic location. For completeness, it should be added, that coastal ocean polynyas are distinctly different from open ocean polynyas in that the former are associated with sea ice production (SIP), salt-rejection and hence DSW formation (Massom et al., 1998; Morales Maqueda et al., 2004; Tamura et al., 2008). This is in contrast to open ocean polynyas (i.e. sensible heat polynyas) that are not typically associated with SIP, especially in the Southern Ocean. Categorically, it is the water mass modification occurring in coastal polynyas of Antarctica which are a key driver of the MOC (Gawarkiewicz and Chapman, 1995; Orsi et al., 1999; Galton-Fenzi et al., 2012; Sansiviero et al., 2017), and particularly, of concern in this work, is that the CDP has been shown to contribute 13–30% of the total AABW mass (Ohshima et al., 2013).

Given the identification of the CDP as a significant contributor to AABW, the effort to determine its controlling factors has been pursued (Nakayama et al., 2014; Liu et al., 2017). Fraser et al. (2019) have shown that the contribution, specifically the length of landfast ice (fast ice), termed

Cape Darnley fast ice (CDFI) is the most corollary physical feature (measured by satellite) in relation to the size of CDP. (Note, I include here the 'dagger' portion, defined by Fraser et al. (2019), when considering CDFI). Beyond the physical importance of fast ice with polynya dynamics it should be further ascribed that their role within the environment as unique habitats is critical and our knowledge is formative (Trathan et al., 2011; Ainley et al., 2015). It may also be worth mentioning here, that conceptually differentiating fast ice from other types of sea ice (e.g. pack ice, floes, etc.), I do so inline with common practice via the World Meteorological Definition (WMO) definition WMO (1970):

> Sea ice which forms and remains fast along the coast, where it is attached to the shore, to an ice wall, to an ice front, between shoals or grounded icebergs. Vertical fluctuations may be observed during changes of sea-level. Fast ice may be formed in situ from sea water or by freezing of floating ice of any age to the shore, and it may extend a few metres or several hundred kilometres from the coast. Fast ice may be more than one year old and may then be prefixed with the appropriate age category (old, second-year, or multi- year).

In many parts of the Antarctic continental shelf bathymetry favours the grounding of icebergs (Wesche and Dierking, 2012). The grounding of icebergs provides a linking/trapping/blocking mechanism for sea ice to become fastened to the iceberg (**?**Fraser et al., 2019; Li et al., 2020). In some cases this fastening links back to the shoreline, shelf edge, grounding keels, glacial tongues, or ice walls (Massom et al., 1998; Fraser et al., 2012). CDFI (and hence CDP) appear connected in this manner to the seafloor topography adjacent to Cape Darnley (i.e. Fram Bank, see 1) (Nakayama et al., 2014; Liu et al., 2017). It has been shown that the spatial and temporal distribution of fast ice acts as barrier and either inhibits or an enhances the size a coastal ocean polynya (Massom et al., 2001; Ohshima et al., 2013; Nakayama et al., 2014; Nihashi and Ohshima, 2015; Kusahara et al., 2017a; Fraser et al., 2019). The work done recently by Fraser et al. (2019) provides a new metric for fast ice categorisation and has shown that quantifying and predicting SIP cannot rely solely on representation of the near surface wind, but also must include the spatial and temporal distribution of the fast ice. Hence, a growing fast ice database of Antarctic is underway (Fraser et al., 2020) and will be utilised in this work.

Prior to the above work, a lack of appropriate repressentation of fast ice as well as the critical iceberg distribution, at a given time, in a coupled ice-ocean numerical model, under represents SIP, and hence DSW formation, incorrectly, or not at all (Nakayama et al., 2014; Liu et al., 2017). Fast–ice estimates derived from satellite indicate the CDP to be a region of high SIP and DSW formation (13–30% of total volume) through annual production of sea ice between 195 ± 71 km^3 (Ohshima et al., 2013; Tamura et al., 2016). The controlling mechanisms for the CDP are yet to be well documented but are likely both atmospheric and oceanographic via: katabatic surface winds, westward circumpolar flow, and favourable bathymetric topology (see 1) of the subsurface northward extension of the CDP.

The impetus of this thesis is focused on correctly representing fast ice via a coupled sea ice–ocean model — the Regional Oceanographic Modelling System (ROMS) (Shchepetkin and McWilliams, 2005), and Los Alamos sea ice model (CICE) (Hunke et al., 2010a,b), which when coupled together are colloquial called `METROMS`. Rationale of choosing these models as well as an overview of regional oceanographic, atmospheric and cryospheric features and concepts will be given in future docusments.

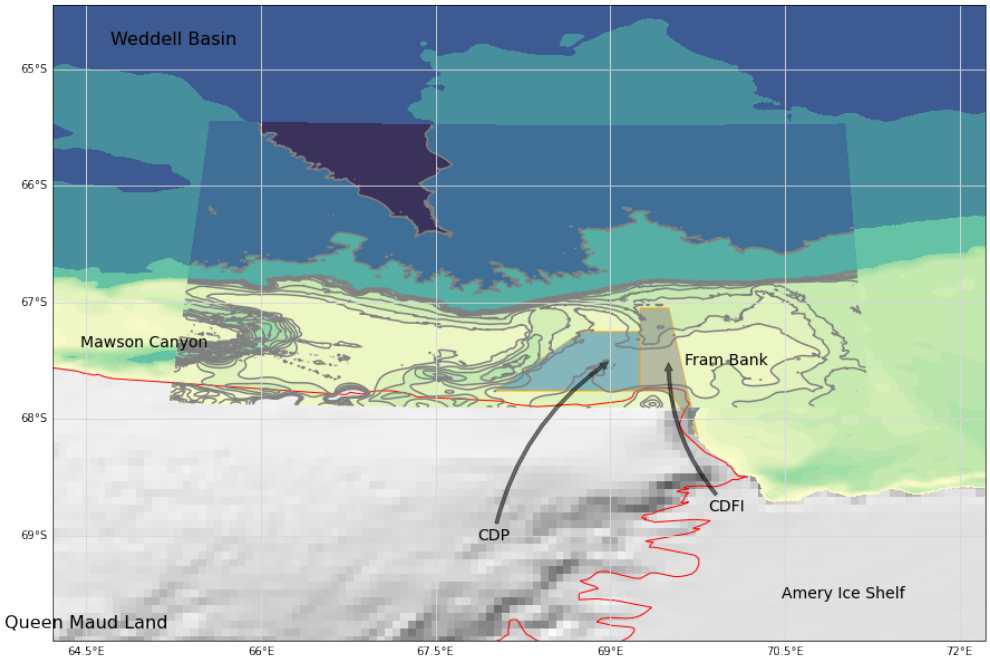


Figure 1: Geoscience Australia bathymetry 2020 dataset (Smith et al., 2020) overlaid on GEBCO and ETOPO1. Cape Darnley Polynya (CDP) and Cape Darnely Fast Ice (CDFI) are roughly indicated as well as as some major geographical features of the region.

Objective

There are three primary objectives of this thesis:

1. **Accurately (spatially and temporally) model fast ice distribution along the Mawson Coast and Cape Darnley region of East Antarctica.**
2. **Quantify the annual rate of DSW from the CDP and hence its mean contribution to AABW. Identification of the controlling mechanism(s) will be highlighted in this objective. This will lead to the tertiary objective, which will be to:**
3. **Assess whether prognostic fast ice is required to accurately model Antarctic shelf water dynamics or if prescribing it is sufficient.**

In addition to these scientific objectives, there will be a range of pedagogical/academic by-products. Namely through the setup and parameterisation of a geophysical fluid dynamic model, coupled sea ice model and then interpretation and validation assessment of those results in the context of a geographically constrained ocean and cryosphere I will gain a deep understanding of physics of these Earth systems.

Methodology

Based on geofluid dynamic principles, using a hydrostatic, edddy-resolving ocean model coupled to sea ice model using tensile strength parameterisation.

1. METROMS build foundational knoledge
   (a) build personal knowledge of ROMS
      i. integrate into ROMS community of users
      ii. compile ROMS on NECTAR
      iii. Australian southeast coast test case:
         A. form an understanding of how to compile ROMS for a real case
         B. form an understanding of to build a curvilinear grid with real bathymetry and coastline
         C. form an understanding of ROMS inputs, setup and parameters
         D. review of model parameters and technical details/underpinnings
         E. run model as both a hindcast and a forecast to build understanding of atmospheric and oceanographic forcing
         F. run simple model validation with ARGO and satellite SST (maybe SSS) that I know will serve me well in using more complex and sparse data in validating EAFIM
         G. stay cautious here in not wanting to do enough work to write-up results
   (b) Construct/justify EAFIM (East Antarctic Fast Ice Model) parameterisation:

i. temporal considerations
   A. total timeframe **must be strategically aligned with validation data**
   B. timesteps must be consistent with capturing dynamic processes; on the order of minutes, but
ii. spatial considerations
   A. domain defintion **must be strategically aligned with validation data**
   B. what horizontal grid resolution will capture the dynamics of polynya and fast-ice
   C. compare available bathymetric sources
   D. bathymetry domain (CDP) bathymetry optimisation
   E. can one be used or is there an optimisation method for using multiple sources
   F. iceberg distribution
   G. leverage-off Van Achter et al. (2021)
   H. have a good look of other possible methods of prescribing icebergs
   I. justify iceberg distribution, while prescribed, is reasonable, it depends on a mean *concentration* of grounded icebergs, rather than actual location . . . is this a well supported assumption?
iii. model forcing
   A. atmospheric forcing
   B. identify which re-analysis and understand spatial limiations ; reference Hout paper once puplished
   C. organise dataset in a format for METROMS to use
   D. oceanographic forcing
   E. identify which re-analysis ocean dataset or circumpolar model to use
   F. organise dataset in a format for METROMS to use
iv. CICE coupling to ROMS
   A. CICE model foundational knowledge
   B. jutify why CICE
   C. learn CICE
   D. learn how to compile with ROMS using `MCT` software
   E. CICE model learning might be a natural follow-on from iceberg distribution

F. likely running a CICE-ROMS test cast for Mawson coast domain will be huge benefit here

G. **CICE fast-ice parameterisation** → this is a key component of my research

2. EAFIM calibration → this is the spin-up required for model stability
   (a) how do I test stability?
3. EAFIM run(s) in order to meet **OBJECTIVES 1 & 2**
4. EAFIM run(s) in order to meet **OBJECTIVES 3**
   (a) NOFI → no fast ice Liu et al. (2017)
   (b) STFI → prescribed fast ice Kusahara et al. (2017b)
   (c) CLFI → prescribed fast ice based on climatology (geometrical shifting on the order of semi-monthly roughly inline with Fraser et al. (2020) dataset)
   (d) PGFI → fully prognostic fast ice, which I'm envisioning as separate to the above runs (i.e. IOT to meet Objective 1&2), however, I don't know at this stage how it will be different.
5. EAFIM validation
   (a) hydrosphere
      i. satellite → SST (Paul has a dataset from the BOM that he has offered)
      ii. MEOP-CTD
      iii. ARGO
      iv. moorings
      v. ship-based CTD
      vi. gliders
      vii. vessels of opportunity
      viii. radar
      ix. XBT
   (b) cryosphere
      i. sea ice concentration
      ii. sea ice thickness
      iii. fast ice distribution using (Fraser et al., 2020)

Outputs

Possible journals:

1. JGR-Oceans
2. The Cryosphere
3. Ocean Modelling

Possible article titles:

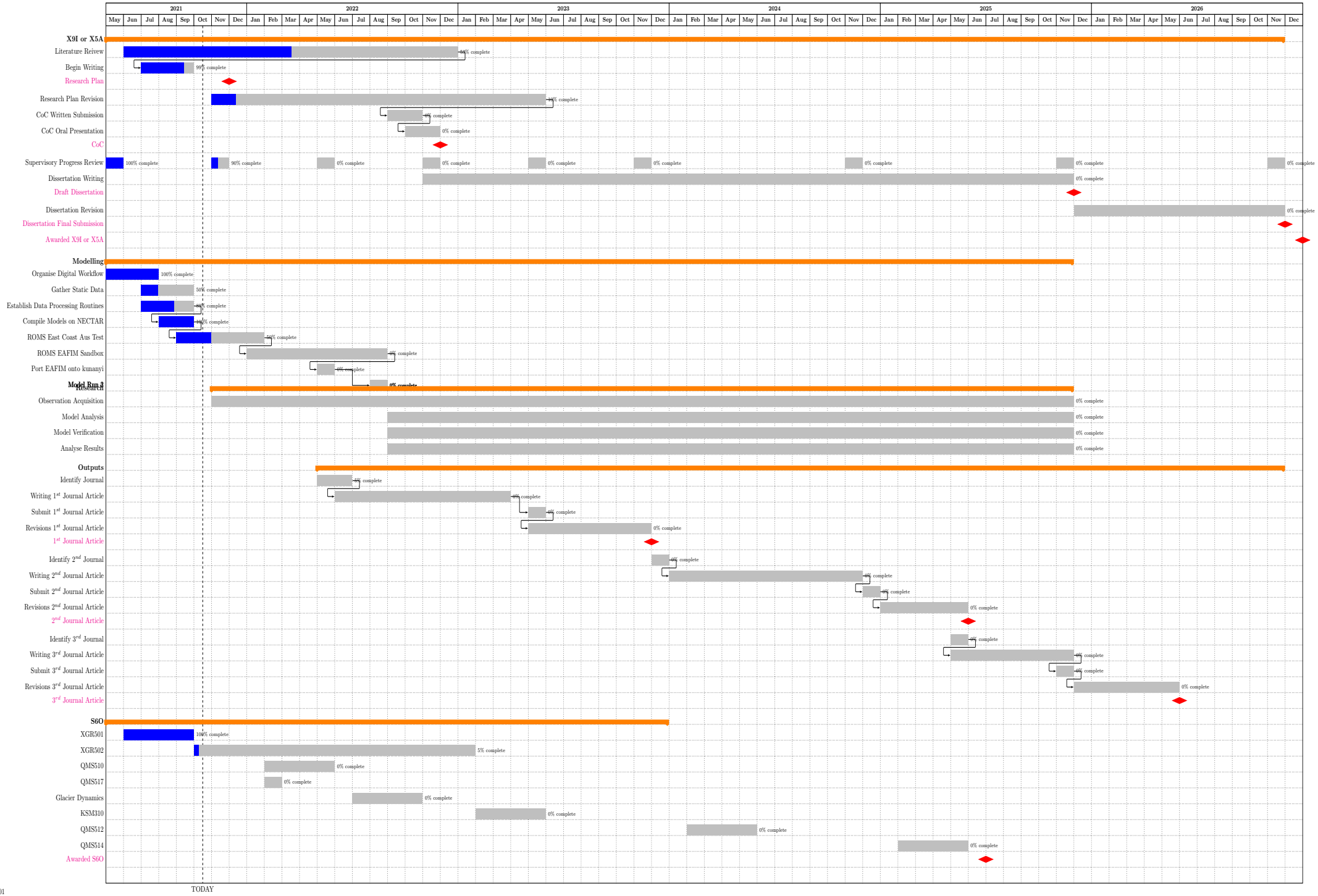
1. East Antarctic Fast Ice Model (EAFIM) – temporal and spatial prognostic accuracy of fast ice annual formation and cessation.
2. Accurate assessment of dense shelf water formation of the Cape Darnley Polynya.
3. Is fast ice required in global models to accurately assess dense shelf water formation?

Possible datasets:

1. mapping of grounded icebergs $10km^2$
2. fast ice review paper (co-author)
3. EAFIM GitHub Branch

Timeline

Please note that I'm a 0.5FTE student planning on completing in 5.5yrs.


2021
2022
2023
2024
2025
2026
May Jun Jul Aug Sep Oct Nov Dec Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec
X9I or X5A
Literature Reivew
Begin Writing
Research Plan
Research Plan Revision
CoC Written Submission
CoC Oral Presentation
CoC
Supervisory Progress Review
Dissertation Writing
Draft Dissertation
Dissertation Revision
Dissertation Final Submission
Awarded X9I or X5A
Modelling
Organise Digital Workflow
Gather Static Data
Establish Data Processing Routines
Compile Models on NECTAR
ROMS East Coast Aus Test
ROMS EAFIM Sandbox
Port EAFIM onto kunanyi
Model Run 3
Research
Observation Acquisition
Model Analysis
Model Verification
Analyse Results
Outputs
Identify Journal
Writing 1st Journal Article
Submit 1st Journal Article
Revisions 1st Journal Article
1st Journal Article
Identify 2nd Journal
Writing 2nd Journal Article
Submit 2nd Journal Article
Revisions 2nd Journal Article
2nd Journal Article
Identify 3rd Journal
Writing 3rd Journal Article
Submit 3rd Journal Article
Revisions 3rd Journal Article
3rd Journal Article
S6O
XGR501
XGR502
QMS510
QMS517
Glacier Dynamics
KSM310
QMS512
QMS514
Awarded S6O
TODAY

2021-10-21-01

Budget

Scholarships

UTas RTP

In receipt as part-time candidate. Essential.

AAPP Top-Up

Application withdrawn until employment level reduces to where this is required. Not essential.

SCAR

Applying for in 2022. Nice for international travel and collaboration.

Travel

Remote to Hobart

Conferences

Collaborative

Infrastucture

Computing Hardware

- Macbook Pro → paid for by IMAS
- iPad

Digital Subscriptions

- Telstra
- Terminus.app
- GoodReader.app

Professional Memberships

- AMOS → student member
- TOS → student member
- AGU → student member
- IGS → student member

Ethical Approvals

Nil requirements.

Work, Health and Safety

Ergonomic risk assessment. Maintain good working posture and frequent breaks. Maintain good diet. Maintain a good exercise regime, at least 60 minutes of walking with body weight routine. Maintain sleep of at least 7 hours per night. Continue with meditation and music playing to relax and keep mentally fit.

Intellectual Property

This work is licensed under a Creative Commons "Attribution-NonCommercial-ShareAlike 3.0 Unported" license.

Project Infrastructure

- NECTAR
- Cloudstor
- kunanyi
- Macbook Pro
  - literature review
  - writing
  - email
  - HPC communication
  - visual meeting
  - code development
- iPad
  - literature review
  - email
  - HPC communication

Skills

- Analytical Physics
- Cryosphere Data Sampling
- Satellite Remote Sensing
- Geodectic Analysis
- Numerical Modelling

- Scientific Computer Programming
- High Performance Computing
- Scientific Writing
- Scientific Presentation

References

References

D. G. Ainley, M. A. Larue, I. Stirling, S. Stammerjohn, and D. B. Siniff. An apparent population decrease, or change in distribution, of Weddell seals along the Victoria Land coast. *Marine Mammal Science*, 31(4):1338–1361, 2015. doi: https://doi.org/10.1111/mms.12220. URL https://onlinelibrary.wiley.com/doi/abs/10.1111/mms.12220.

S. Aoki, K. Katsumata, M. Hamaguchi, A. Noda, Y. Kitade, K. Shimada, D. Hirano, D. Simizu, Y. Aoyama, K. Doi, and Y. Nogi. Freshening of Antarctic Bottom Water off Cape Darnley, East Antarctica. *Journal of Geophysical Research: Oceans*, 125(8):e2020JC016374, 2020. doi: https://doi.org/10.1029/2020JC016374. URL https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1029/2020JC016374. e2020JC016374 2020JC016374.

D. Barber and R. Massom. Chapter 1: The role of sea ice in Arctic and Antarctic polynyas. *Elsevier oceanography series*, 74:1–54, 2007.

N. Bindoff, J. Willebrand, V. Artale, A. Cazenave, J. Gregory, S. Gulev, K. Hanawa, C. Le Quéré, S. Levitus, Y. Nojiri, C. Shum, L. Talley, and U. Alakkat. Observations: Oceanic climate change and sea level. *The Physical Science Basis. Working Group I Contribution to the Intergovernmental Panel on Climate Change Fourth Assessment Report*, 01 2007.

J. Boé, A. Hall, and X. Qu. Deep ocean heat uptake as a major source of spread in transient climate change simulations. *Geophysical Research Letters*, 36(22), 2009. doi: https://doi.org/10.1029/2009GL040845. URL https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1029/2009GL040845.

C. M. Domingues, J. A. Church, N. J. White, P. J. Gleckler, S. E. Wijffels, P. M. Barker, and J. R. Dunn. Improved estimates of upper-ocean warming and multi-decadal sea-level rise. *Nature*, 453(7198):1090–1093, 2008. doi: 10.1038/nature07080. URL https://doi.org/10.1038/nature07080.

P. J. Durack, P. J. Gleckler, S. G. Purkey, G. C. Johnson, and T. P. Boyer. Ocean warming: From the surface to the deep in observations and models. *Oceanography*, 31(2):41–51, June 2018. URL https://doi.org/10.5670/oceanog.2018.227.

A. D. Fraser, R. A. Massom, K. J. Michael, B. K. Galton-Fenzi, and J. L. Lieser. East Antarctic landfast sea ice distribution and variability, 2000–08. *Journal of Climate*, 25(4):1137 – 1156, 2012. doi: 10.1175/JCLI-D-10-05032.1. URL https://journals.ametsoc.org/view/journals/clim/25/4/jcli-d-10-05032.1.xml.

A. D. Fraser, K. I. Ohshima, S. Nihashi, R. A. Massom, T. Tamura, K. Nakata, G. D. Williams, S. Carpentier, and S. Willmes. Landfast ice controls on sea-ice production in the Cape Darnley

Polynya: A case study. *Remote Sensing of Environment*, 233:111315, 2019. ISSN 0034-4257. doi: https://doi.org/10.1016/j.rse.2019.111315. URL https://www.sciencedirect.com/science/article/pii/S0034425719303347.

A. D. Fraser, R. A. Massom, K. I. Ohshima, S. Willmes, P. J. Kappes, J. Cartwright, and R. Porter-Smith. High-resolution mapping of circum-Antarctic landfast sea ice distribution, 2000–2018. *Earth System Science Data*, 12(4):2987–2999, 2020. doi: 10.5194/essd-12-2987-2020. URL https://essd.copernicus.org/articles/12/2987/2020/.

B. K. Galton-Fenzi, J. R. Hunter, R. Coleman, S. J. Marsland, and R. C. Warner. Modeling the basal melting and marine ice accretion of the Amery Ice Shelf. *Journal of Geophysical Research: Oceans*, 117(C9), 2012. doi: https://doi.org/10.1029/2012JC008214. URL https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1029/2012JC008214.

G. Gawarkiewicz and D. C. Chapman. A numerical study of dense water formation and transport on a shallow, sloping continental shelf. *Journal of Geophysical Research: Oceans*, 100(C3):4489–4507, 1995. doi: https://doi.org/10.1029/94JC01742. URL https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1029/94JC01742.

E. C. Hunke, W. H. Lipscomb, A. Turner, N. Jeffery, and S. Elliott. CICE: the Los Alamos sea ice gmodel documentation and software user's manual version 4.1. *Los Alamos National Laboratory, Los Alamos, NM*, pages 1–115, 2010a.

E. C. Hunke, W. H. Lipscomb, and A. K. Turner. Sea-ice models for climate study: retrospective and new directions. *Journal of Glaciology*, 56(200):1162–1172, 2010b. doi: 10.3189/002214311796406095.

G. C. Johnson. Quantifying Antarctic Bottom Water and North Atlantic Deep Water volumes. *Journal of Geophysical Research: Oceans*, 113(C5), 2008. doi: https://doi.org/10.1029/2007JC004477. URL https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1029/2007JC004477.

K. Kusahara, H. Hasumi, A. D. Fraser, S. Aoki, K. Shimada, G. D. Williams, R. Massom, and T. Tamura. Modeling ocean–cryosphere interactions off Adélie and George V Land, East Antarctica. *Journal of Climate*, 30(1):163 – 188, 2017a. doi: 10.1175/JCLI-D-15-0808.1. URL https://journals.ametsoc.org/view/journals/clim/30/1/jcli-d-15-0808.1.xml.

K. Kusahara, G. D. Williams, T. Tamura, R. Massom, and H. Hasumi. Dense shelf water spreading from Antarctic coastal polynyas to the deep Southern Ocean: A regional circumpolar model study. *Journal of Geophysical Research: Oceans*, 122(8):6238–6253, 2017b. doi: https://doi.org/10.1002/2017JC012911. URL https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1002/2017JC012911.

S. Levitus, J. Antonov, and T. Boyer. Warming of the world ocean, 1955–2003. *Geophysical Research Letters*, 32(2), 2005. doi: https://doi.org/10.1029/2004GL021592. URL https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1029/2004GL021592.

X. Li, M. Shokr, F. Hui, Z. Chi, P. Heil, Z. Chen, Y. Yu, M. Zhai, and X. Cheng. The spatio-temporal patterns of landfast ice in Antarctica during 2006–2011 and 2016–2017 using high-resolution SAR imagery. *Remote Sensing of Environment*, 242:111736, 2020. ISSN 0034-4257. doi:

https://doi.org/10.1016/j.rse.2020.111736. URL https://www.sciencedirect.com/science/article/pii/S0034425720301061.

C. Liu, Z. Wang, C. Cheng, R. Xia, B. Li, and Z. Xie. Modeling modified Circumpolar Deep Water intrusions onto the Prydz Bay continental shelf, East Antarctica. *Journal of Geophysical Research: Oceans*, 122(7):5198–5217, 2017. doi: https://doi.org/10.1002/2016JC012336. URL https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1002/2016JC012336.

R. A. Massom, P. Harris, K. J. Michael, and M. Potter. The distribution and formative processes of latent-heat polynyas in East Antarctica. *Annals of Glaciology*, 27:420–426, 1998. doi: 10.3189/1998AoG27-1-420-426.

R. A. Massom, K. L. Hill, V. I. Lytle, A. P. Worby, M. Paget, and I. Allison. Effects of regional fast-ice and iceberg distributions on the behaviour of the Mertz Glacier polynya, East Antarctica. *Annals of Glaciology*, 33:391–398, 2001. doi: 10.3189/172756401781818518.

A. Meijers, A. Klocker, N. Bindoff, G. Williams, and S. Marsland. The circulation and water masses of the Antarctic shelf and continental slope between 30 and 80°E. *Deep Sea Research Part II: Topical Studies in Oceanography*, 57(9):723–737, 2010. ISSN 0967-0645. doi: https://doi.org/10.1016/j.dsr2.2009.04.019. URL https://www.sciencedirect.com/science/article/pii/S0967064509003865. "BROKE-West" a Biological/Oceanographic Survey Off the Coast of East Antarctica (30-80E) Carried Out in January-March 2006.

G. Mizuta, Y. Fukamachi, D. Simizu, Y. Matsumura, Y. Kitade, D. Hirano, M. Fujii, Y. Nogi, and K. I. Ohshima. Seasonal evolution of Cape Darnley Bottom Water revealed by mooring measurements. *Frontiers in Marine Science*, 8:1054, 2021. ISSN 2296-7745. doi: 10.3389/fmars.2021.657119. URL https://www.frontiersin.org/article/10.3389/fmars.2021.657119.

M. A. Morales Maqueda, A. J. Willmott, and N. R. T. Biggs. Polynya dynamics: a review of observations and modeling. *Reviews of Geophysics*, 42(1), 2004. doi: https://doi.org/10.1029/2002RG000116. URL https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1029/2002RG000116.

D. M. Murphy, S. Solomon, R. W. Portmann, K. H. Rosenlof, P. M. Forster, and T. Wong. An observationally based energy balance for the earth since 1950. *Journal of Geophysical Research: Atmospheres*, 114(D17), 2009. doi: https://doi.org/10.1029/2009JD012105. URL https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1029/2009JD012105.

Y. Nakayama, K. I. Ohshima, Y. Matsumura, Y. Fukamachi, and H. Hasumi. A numerical investigation of formation and variability of Antarctic Bottom Water off Cape Darnley, East Antarctica. *Journal of Physical Oceanography*, 44(11):2921 – 2937, 2014. doi: 10.1175/JPO-D-14-0069.1. URL https://journals.ametsoc.org/view/journals/phoc/44/11/jpo-d-14-0069.1.xml.

K. A. Naughten, K. J. Meissner, B. K. Galton-Fenzi, M. H. England, R. Timmermann, H. H. Hellmer, T. Hattermann, and J. B. Debernard. Intercomparison of Antarctic ice-shelf, ocean, and sea-ice interactions simulated by MetROMS-iceshelf and FESOM 1.4. *Geoscientific Model Development*, 11(4):1257–1292, 2018. doi: 10.5194/gmd-11-1257-2018. URL https://gmd.copernicus.org/articles/11/1257/2018/.

S. Nihashi and K. I. Ohshima. Circumpolar mapping of Antarctic coastal polynyas and landfast sea ice: Relationship and variability. *Journal of Climate*, 28(9):3650 – 3670, 2015. doi: 10.

1175/JCLI-D-14-00369.1. URL https://journals.ametsoc.org/view/journals/clim/28/9/jcli-d-14-00369.1.xml.

K. I. Ohshima, Y. Fukamachi, G. D. Williams, S. Nihashi, F. Roquet, Y. Kitade, T. Tamura, D. Hirano, L. Herraiz-Borreguero, I. Field, M. Hindell, S. Aoki, and M. Wakatsuchi. Antarctic Bottom Water production by intense sea-ice formation in the Cape Darnley Polynya. *Nature Geoscience*, 6(3):235–240, Mar 2013.

K. I. Ohshima, S. Nihashi, and K. Iwamoto. Global view of sea-ice production in polynyas and its linkage to dense/bottom water formation. *Geoscience Letters*, 3(1):13, 2016. doi: 10.1186/s40562-016-0045-4. URL https://doi.org/10.1186/s40562-016-0045-4.

A. Orsi, G. Johnson, and J. Bullister. Circulation, mixing, and production of Antarctic Bottom Water. *Progress in Oceanography*, 43(1):55–109, 1999. ISSN 0079-6611. doi: https://doi.org/10.1016/S0079-6611(99)00004-X. URL https://www.sciencedirect.com/science/article/pii/S007966119900004X.

A. H. Orsi, S. S. Jacobs, A. L. Gordon, and M. Visbeck. Cooling and ventilating the abyssal ocean. *Geophysical Research Letters*, 28(15):2923–2926, 2001. doi: https://doi.org/10.1029/2001GL012830. URL https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1029/2001GL012830.

H.-O. Pörtner, D. Roberts, V. Masson-Delmotte, P. Zhai, M. Tignor, E. Poloczanska, K. Mintenbeck, A. Alegría, M. Nicolai, A. Okem, J. Petzold, B. Rama, and N. Weyer. IPCC special report on the ocean and cryosphere in a changing climate. In *IPCC Special Report on the Ocean and Cryosphere in a Changing Climate*. IPCC, 2019.

S. G. Purkey and G. C. Johnson. Warming of global abyssal and deep Southern Ocean waters between the 1990s and 2000s: Contributions to global heat and sea level rise budgets. *Journal of Climate*, 23(23):6336–6351, dec 2010. doi: 10.1175/2010jcli3682.1. URL https://doi.org/10.1175/2010jcli3682.1.

S. G. Purkey and G. C. Johnson. Global contraction of Antarctic Bottom Water between the 1980s and 2000s. *Journal of Climate*, 25(17):5830–5844, mar 2012. doi: 10.1175/jcli-d-11-00612.1. URL https://doi.org/10.1175/jcli-d-11-00612.1.

S. R. Rintoul. The global influence of localized dynamics in the Southern Ocean. *Nature*, 558(7709):209–218, 2018. doi: 10.1038/s41586-018-0182-3. URL https://doi.org/10.1038/s41586-018-0182-3.

M. Sansiviero, M. Morales Maqueda, G. Fusco, G. Aulicino, D. Flocco, and G. Budillon. Modelling sea ice formation in the Terra Nova Bay polynya. *Journal of Marine Systems*, 166:4–25, 2017. ISSN 0924-7963. doi: https://doi.org/10.1016/j.jmarsys.2016.06.013. URL https://www.sciencedirect.com/science/article/pii/S092479631630166X. Mesoscale and high-frequency variability in the Ross Sea (Antarctica).

A. F. Shchepetkin and J. C. McWilliams. The regional oceanic modeling system (ROMS): a split-explicit, free-surface, topography-following-coordinate oceanic model. *Ocean Modelling*, 9(4):347–404, 2005. ISSN 1463-5003. doi: https://doi.org/10.1016/j.ocemod.2004.08.002. URL https://www.sciencedirect.com/science/article/pii/S1463500304000484.

J. Smith, Nogi, S. Y., M., B. Dorschel, and A. Leventer. Cape Darnley Bathymetry grid v.1. Bathymetric Grid AU/NZS ISO 19115-1:2014, ISO 19115-1:2014, ISO 19115-3 (Draft Schemas 2015), Geoscience Australia, Canberra, Australia, 2020.

S. D. Smith, R. D. Muench, and C. H. Pease. Polynyas and leads: An overview of physical processes and environment. *Journal of Geophysical Research: Oceans*, 95(C6):9461–9479, 1990. doi: https://doi.org/10.1029/JC095iC06p09461. URL https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1029/JC095iC06p09461.

A. L. Stewart, A. Klocker, and D. Menemenlis. Acceleration and overturning of the Antarctic Slope Current by winds, eddies, and tides. *Journal of Physical Oceanography*, 49(8):2043 – 2074, 2019. doi: 10.1175/JPO-D-18-0221.1. URL https://journals.ametsoc.org/view/journals/phoc/49/8/jpo-d-18-0221.1.xml.

L. D. Talley. Closure of the Global Overturning Circulation Through the Indian, Pacific, and Southern Oceans: Schematics and Transports. *Oceanography*, issue$_volume, March$2013. URL. The overturning pathways for the surface-ventilated North Atlantic Deep Water (NADW) and Antarctic Bottom Water (AABW) and the diffusively formed Indian Deep Water (IDW) and Pacific Deep Water (PDW) are intertwined. The global overturning circulation (GOC) includes both large wind-driven upwelling in the Southern Ocean and important internal diapycnal transformation in the deep Indian and Pacific Oceans. All three northern-source Deep Waters (NADW, IDW, PDW) move southward and upwell in the Southern Ocean. AABW is produced from the denser, salty NADW and a portion of the lighter, low oxygen IDW/PDW that upwells above and north of NADW. The remaining upwelled IDW/PDW stays near the surface, moving into the subtropical thermoclines, and ultimately sources about one-third of the NADW. Another third of the NADW comes from AABW upwelling in the Atlantic. The remaining third comes from AABW upwelling to the thermocline in the Indian-Pacific. Atlantic cooling associated with NADW formation (0.3 PW north of 32°S; 1 PW = 1015 W) and Southern Ocean cooling associated with AABW formation (0.4 PW south of 32°S) are balanced mostly by 0.6 PW of deep diffusive heating in the Indian and Pacific Oceans; only 0.1 PW is gained at the surface in the Southern Ocean. Thus, while an adiabatic model of NADW global overturning driven by winds in the Southern Ocean, with buoyancy added only at the surface in the Southern Ocean, is a useful dynamical idealization, the associated heat changes require full participation of the diffusive Indian and Pacific Oceans, with a basin-averaged diffusivity on the order of the Munk value of 10–4 m2 s–1. .

T. Tamura, K. I. Ohshima, and S. Nihashi. Mapping of sea ice production for Antarctic coastal polynyas. *Geophysical Research Letters*, 35(7), 2008. https://doi.org/10.1029/2007GL032903. URL https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1029/2007GL032903.

T. Tamura, K. I. Ohshima, A. D. Fraser, and G. D. Williams. Sea ice production variability in Antarctic coastal polynyas. *Journal of Geophysical Research: Oceans*, 121(5):2967–2979, 2016. https://doi.orpppg/10.1002/2015JC011537. URL https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1002/2015JC011537.

P. N. Trathan, P. T. Fretwell, and B. Stonehouse. First recorded loss of an Emperor Penguin colony in the recent period of Antarctic regional warming: Implications for other colonies. *PLOS ONE*, 6(2):1–11, 02 2011. 10.1371/journal.pone.0014738. URL https://doi.org/10.1371/journal.pone.0014738.

G. Van Achter, T. Fichefet, H. Goosse, C. Pelletier, J. Sterlin, P.-V. Huot, J.-F. Lemieux, A. Fraser, R. Porter-Smith, and K. Haubner. Modelling landfast sea ice and its influence on ocean-ice interactions in the area of the Totten Glacier, East Antarctica. EGU21-8846, April 2021.

C. Wesche and W. Dierking. Iceberg signatures and detection in SAR images in two test regions of the Weddell Sea, Antarctica. *Journal of Glaciology*, 58(208):325–339, 2012. 10.3189/2012J0G11J020.

G. Williams, S. Nicol, S. Aoki, A. Meijers, N. Bindoff, Y. Iijima, S. Marsland, and A. Klocker. Surface oceanography of BROKE-West, along the Antarctic margin of the south-west Indian Ocean (30–80°E). *Deep Sea Research Part II: Topical Studies in Oceanography*, 57(9):738–757, 2010. ISSN 0967-0645. https://doi.org/10.1016/j.dsr2.2009.04.020. URL https://www.sciencedirect.com/science/article/pii/S0967064509003877. "BROKE-West" a Biological/Oceanographic Survey Off the Coast of East Antarctica (30-80°E) Carried Out in January-March 2006.

G. D. Williams, L. Herraiz-Borreguero, F. Roquet, T. Tamura, K. I. Ohshima, Y. Fukamachi, A. D. Fraser, L. Gao, H. Chen, C. R. McMahon, R. Harcourt, and M. Hindell. The suppression of Antarctic Bottom W]ater formation by melting ice shelves in Prydz Bay, url = https://doi.org/10.1038/ncomms12577, volume = 7, year = 2016, bdsk-file-1 = YnBsaXN0MDDSAQIDBFxyZW bdsk-url-1 = https://doi.org/10.1038/ncomms12577. *Nature Communications*, (1):12577, Aug . ISSN 2041-1723. 10.1038/ncomms12577.

WMO, editor. *WMO sea-ice nomenclature, terminology, codes and illustrated glossary*, volume 1-4, page 147. World Meteorological Organisation, 1970.